

The Three Phases of the Simulation (State Machine)

To make the simulation interactive and strategic, the application should operate in three distinct phases:

Phase 1: Setup & Planning (Active Pause)

- The simulation time is strictly paused (`is_running = False`).
- The user is presented with the initial forest map and a strict resource budget (e.g., 20 firebreaks/barriers).
- The user clicks on the map to place barriers strategically. The GUI must track the remaining budget and prevent the user from placing more barriers than allowed.

Phase 2: The Ignition Point

- Once the defenses are deployed, the user switches the interaction mode (e.g., from "Build" to "Ignite").
- The user clicks on one or more trees to set the starting points of the wildfire. (Alternatively, this can be loaded from a pre-defined JSON scenario).

Phase 3: Execution & Simulation

- The user clicks the "**Start Simulation**" button.
- **Input Lock:** Manual controls (placing barriers or starting fires) are immediately disabled so the user cannot cheat while the fire is spreading.
- The cellular automaton engine takes over. Fire spreads according to the probability rules and the chosen neighborhood (von Neumann/Moore).
- Once the fire naturally extinguishes (no active burning cells left), the system calculates the final metrics and displays the end-game score (e.g., "*Trees Saved: 65%*").

Forest Fire Defense

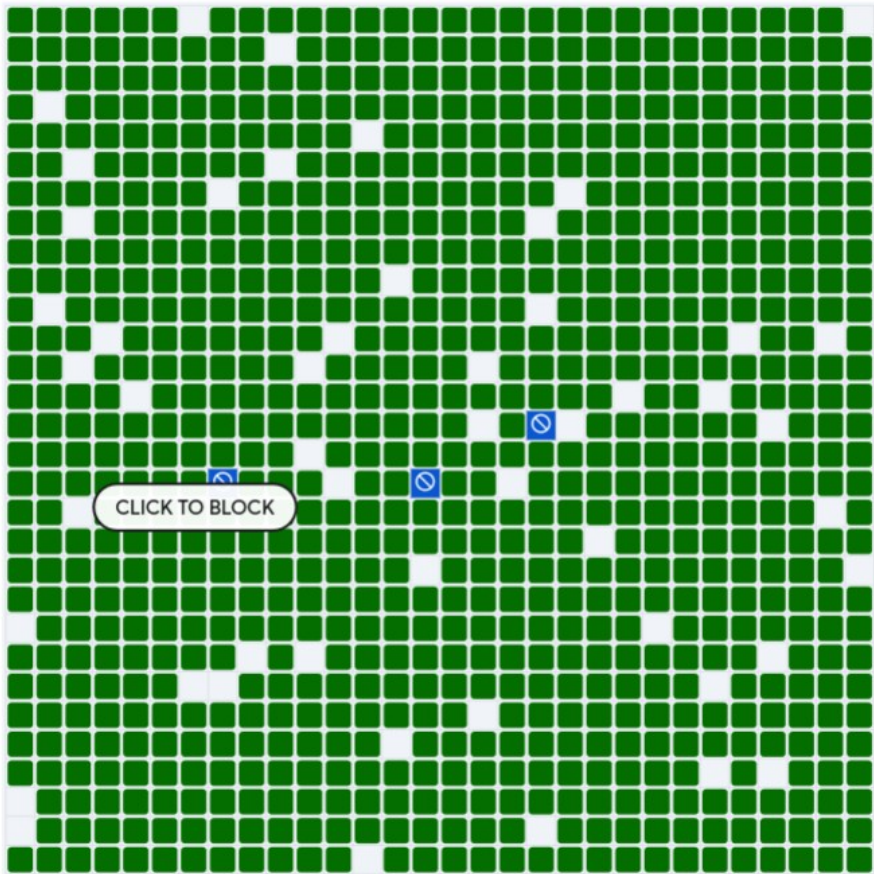
BARRIER BUDGET

TREES SAVED

17

100.0%

Setup Phase: Place barriers and ignite a starting fire.



Action Mode

Build Barriers

Start Simulation

Reset Map

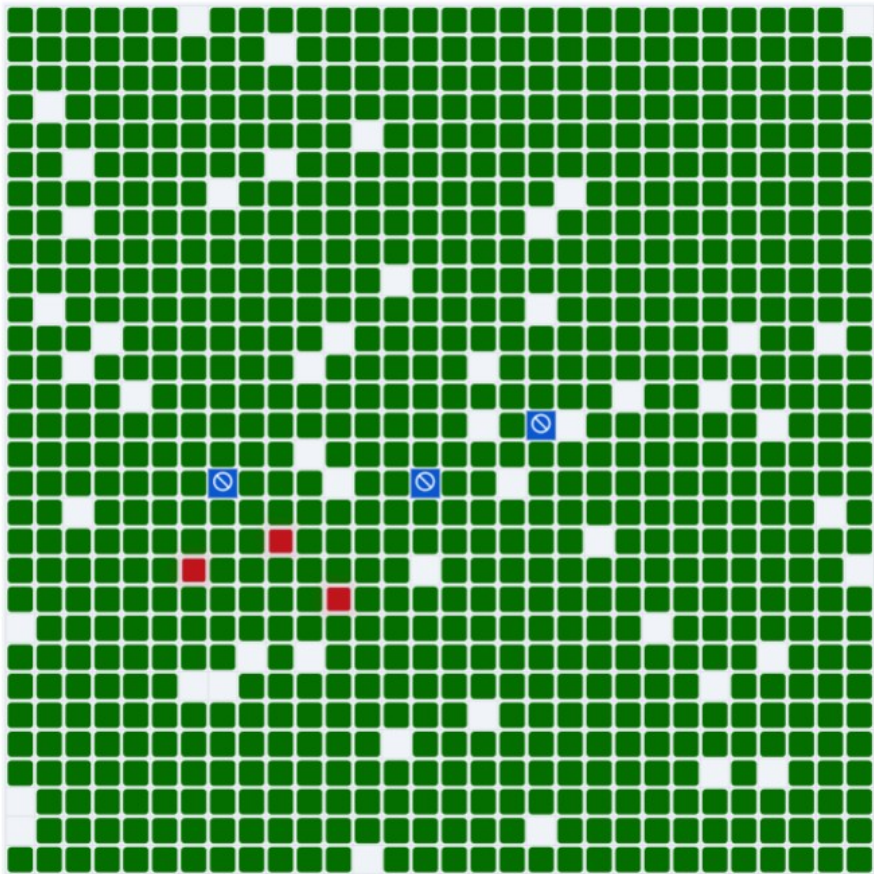
Forest Fire Defense

BARRIER BUDGET
17

TREES SAVED
100.0%

Setup Phase: Place barriers and ignite a starting fire.

CLICK TO BURN



Action Mode

Ignite Fire

Build Barriers

Ignite Fire

Start Simulation

at Map